

Advanced NLP

Session 9 – Scaling laws & In-Context Learning

Teacher: Dr. Behrouz Minaei-Bidgoli
Presenter: Setare Kahnamuee
Presentation creator: Setare Kahnamuee

Iran University of Science & Technology

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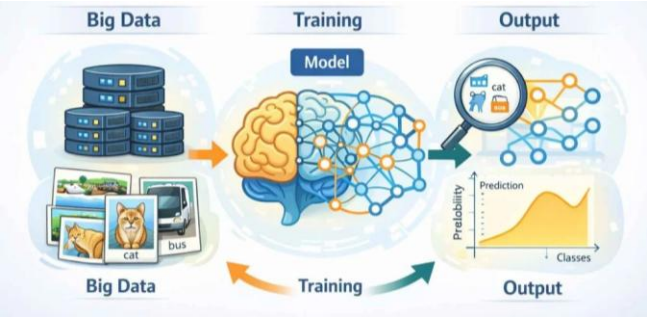
1. Scaling is all you need
2. Scaling laws for data and model size
3. Scaling laws for model engineering
4. In Context Learning
5. Is Scaling all we need?

1.

Scaling is all you need

Main Resource For this Section: https://cs336.stanford.edu/ and https://cmu-l3.github.io/anlp-spring2026

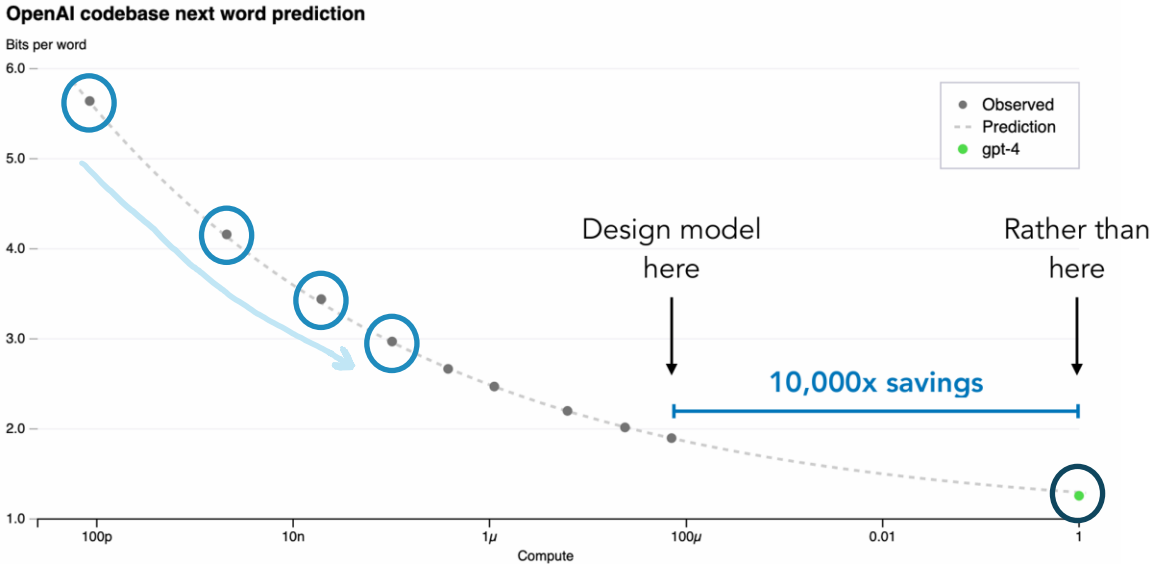
Not Our First Rodeo

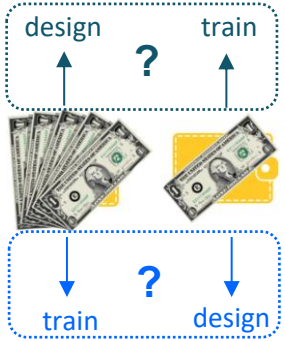




Scaling Laws help us make decisions

Old and unpleasant: tune hyperparameters on big models
New (over?) optimism: tune on small models, extrapolate to large ones





1. Train a few smaller models
2. Establish a scaling law
3. Select based on the scaling law prediction

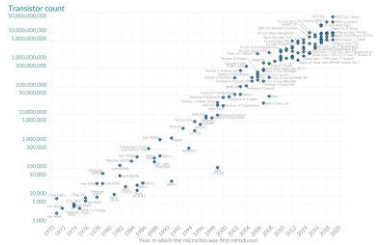
A predictable relationship between two quantities

For additional information: 1, 2

Scale Wars: A debate

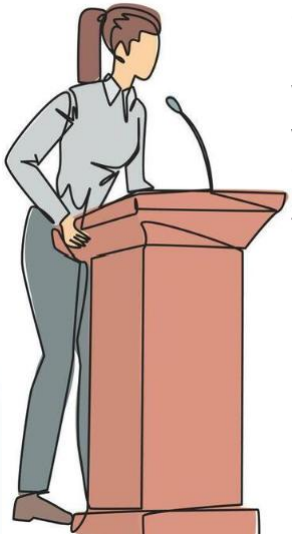
Moore’s Law: The number of transistors on microchips doubles even two years

A successful scaling trend that eventually encountered physical and economic limitations.



"Scaling Works”

Roadmap to future AI progress



- **"The results speak for themselves!"**

Why abandon something that works?

- **"We can predict the future!"**

Scaling laws let us estimate how much better larger models will perform.

"But For How Long?“

Just a description of what has worked so far

- **"So did Moore's Law!"**

Decades of success didn't make Moore's Law last forever. Why assume AI scaling will?

- **"You can't predict forever!"**

An empirical trend isn't a physical law. Extrapolating beyond the evidence is risky.



For additional information: 1, 2, 3

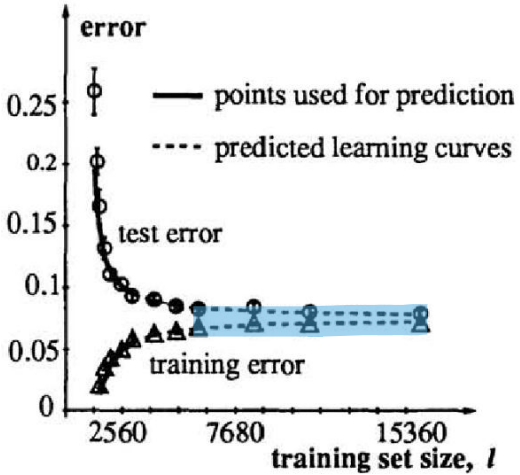
Early history of scaling laws - 1993

Learning Curves: Asymptotic Values and Rate of Convergence

Corinna Cortes, L. D. Jackel, Sara A. Solla, Vladimir Vapnik, and John S. Denker
AT&T Bell Laboratories
Holmdel, NJ 07733

Abstract

Training classifiers on large databases is computationally demanding. It is desirable to develop efficient procedures for a reliable prediction of a classifier's suitability for implementing a given task, so that resources can be assigned to the most promising candidates or freed for exploring new classifier candidates. We propose such a practical and principled predictive method. Practical because it avoids the costly procedure of training poor classifiers on the whole training set, and principled because of its theoretical foundation.



Early history of scaling laws

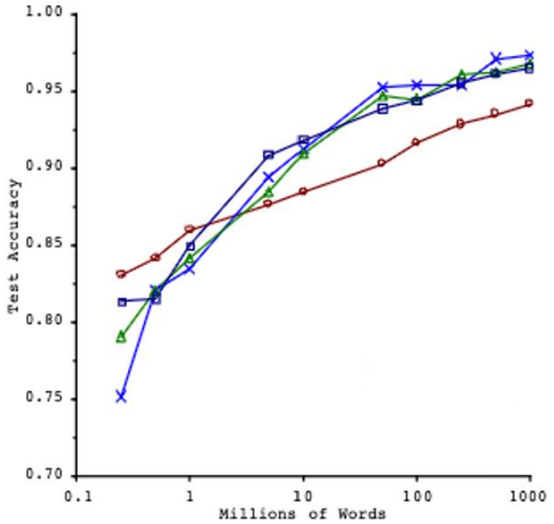
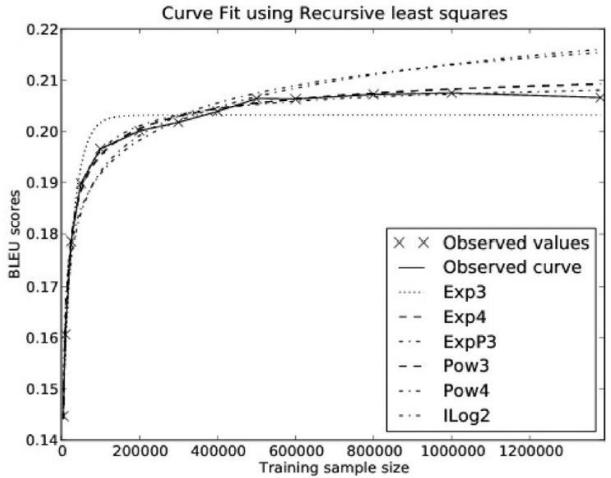


Figure 1. Learning Curves for Confusion Set Disambiguation

2001

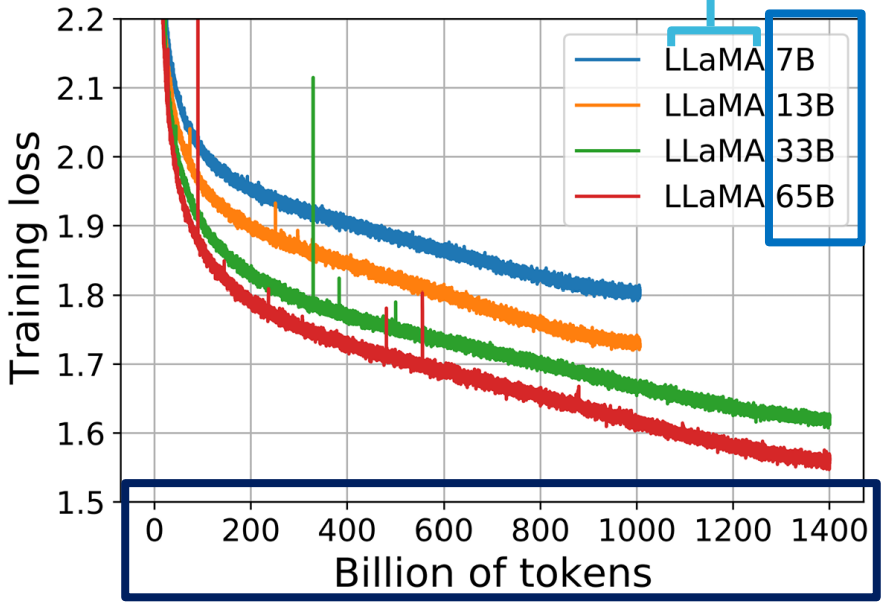


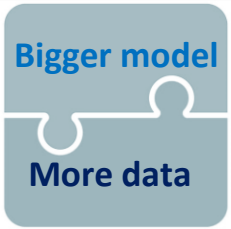
2012

Similar rate? → corpus development > algorithm development

A resource allocation problem

Given a fixed pretraining budget, how should we allocate the budget to achieve the best possible model?





Simple and predictive rules

We spend compute by performing forward and backward passes on training sequences

An approximation for transformer language models:

$$C \approx ND$$

- Number of model parameters
- Number of tokens
- Compute; floating point operations(FLOPs)

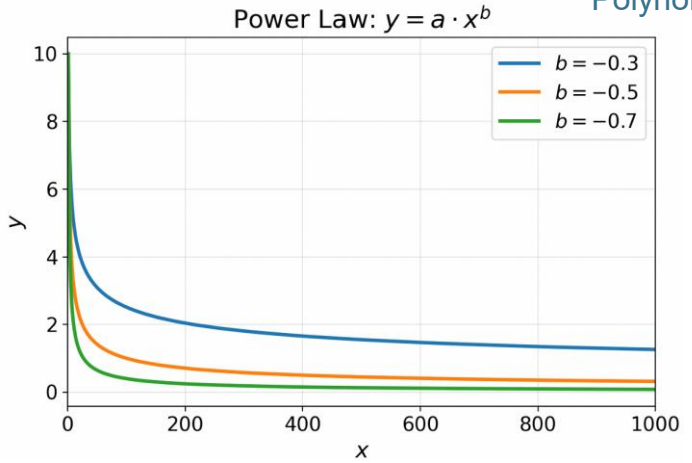
Effect on loss

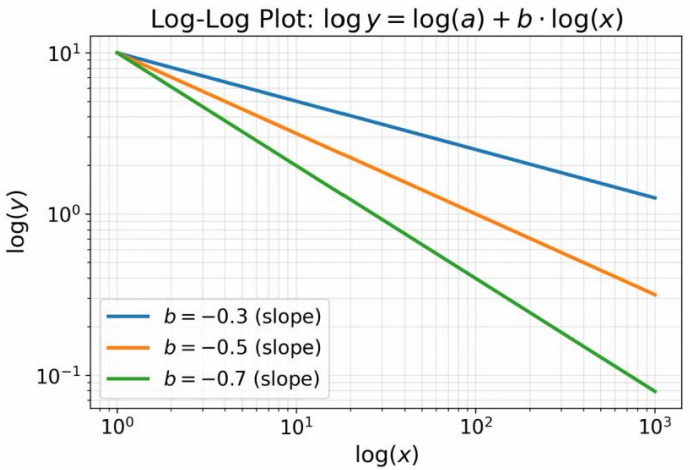
Model size scaling
Data scaling
Compute scaling

Scaling laws

Power law relationships

Polynomial decay





$$y = a \cdot x^b \xrightarrow{\log()} \log y = \log(a) + b \cdot \log(x)$$

Linear on a log-log plot

Log-log plots

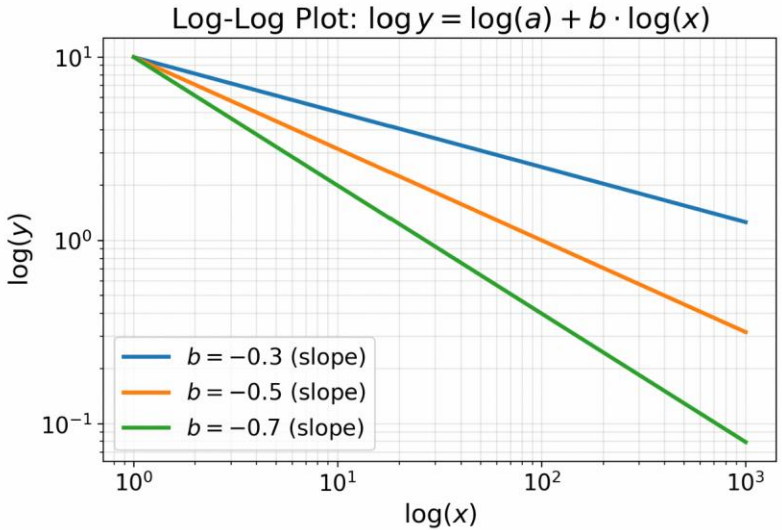
$$y = a \cdot x^b$$

Double scaling

$$y_{\text{new}} = a(2x)^b$$

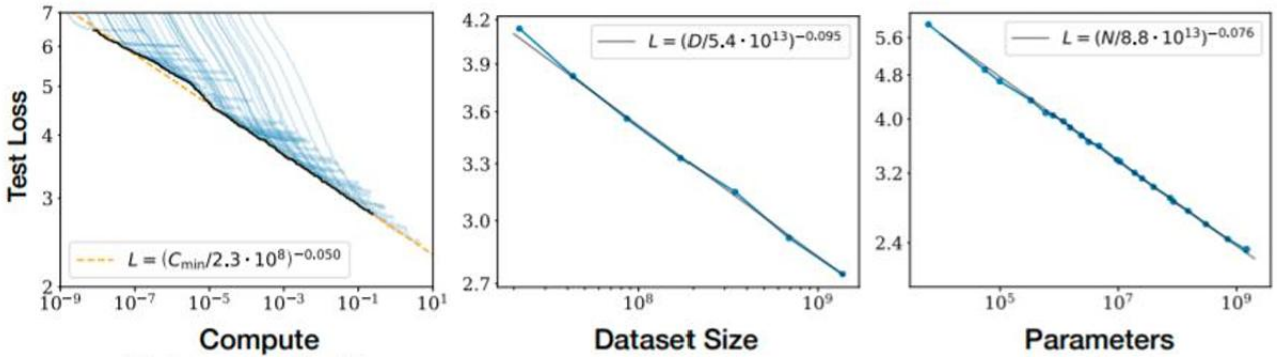
$$= ax^b 2^b$$

$$\Rightarrow y_{\text{new}} = y2^b$$

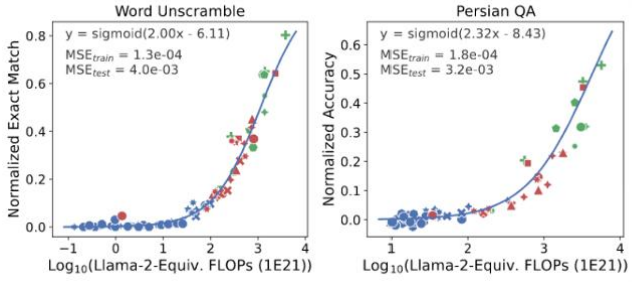


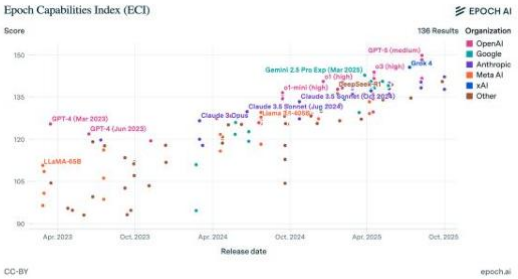
b	$y \to 2^b \cdot y$
-0.3	$y \to 0.81y$ (19% reduction)
-0.5	$y \to 0.71y$ (29% reduction)
-0.7	$y \to 0.62y$ (38% reduction)

Scaling law relationships for many factors



The usual ones





Downstream tasks

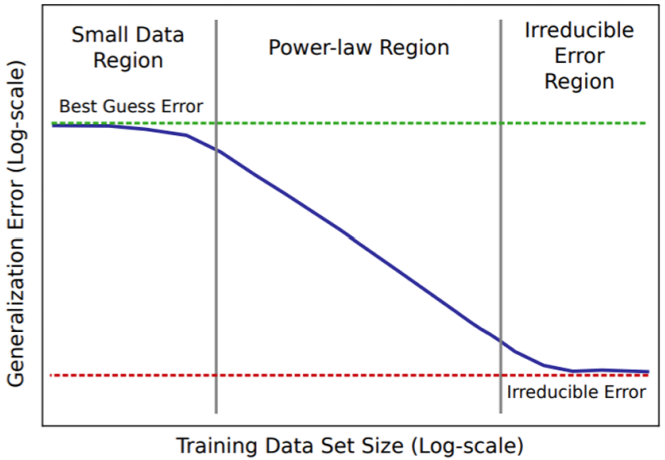
2.

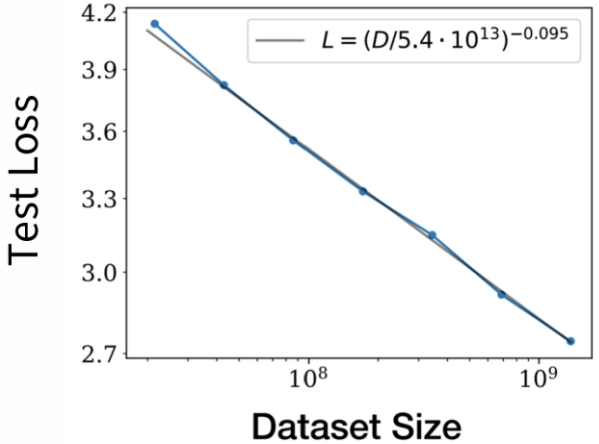
Scaling laws for data and model size

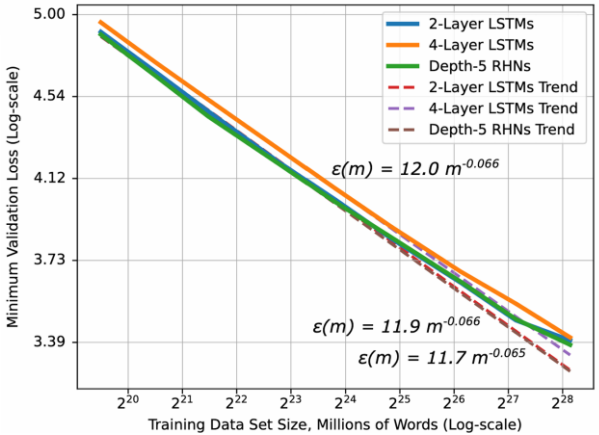
Main Resource For this Section: https://cs336.stanford.edu/ and https://cmu-l3.github.io/anlp-spring2026

Data scaling law

- Ensure model size is not bottlenecked → large N
- Train with varied dataset size
- Plot the empirical observation
- More data → do better → error decreases



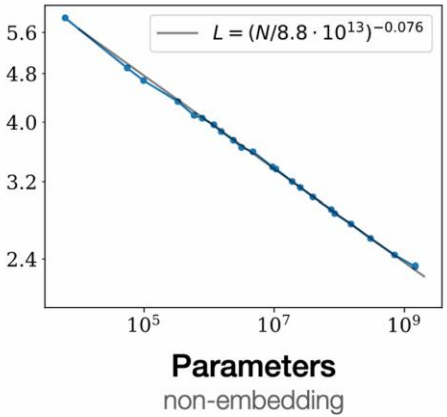


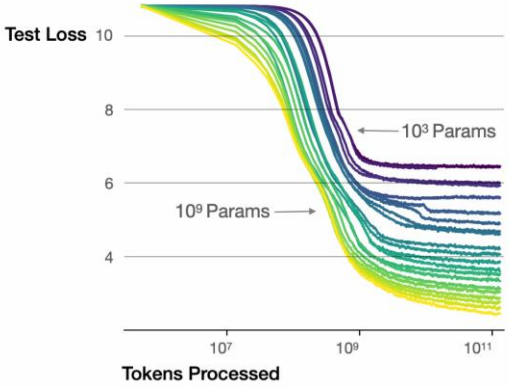


Both follow the trend very well

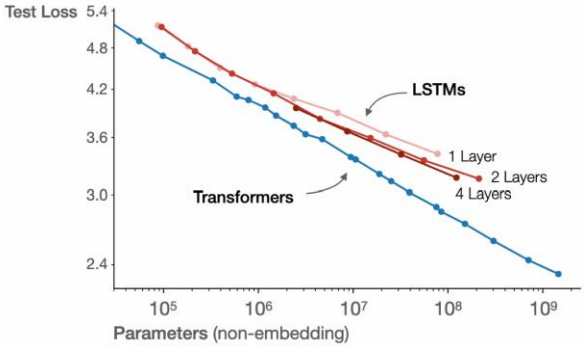
Model size scaling law

Similar relationships observed for model size





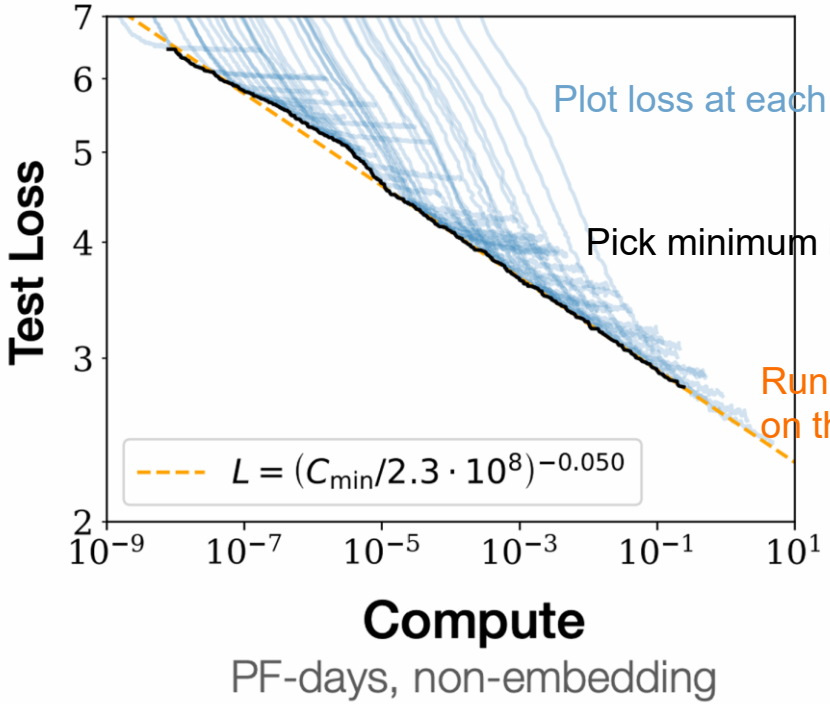
Useful for determining model hyperparameters



Large models require fewer samples to reach the same performance

Compute scaling law

Train models of different sizes and numbers of tokens



Plot loss at each step of training

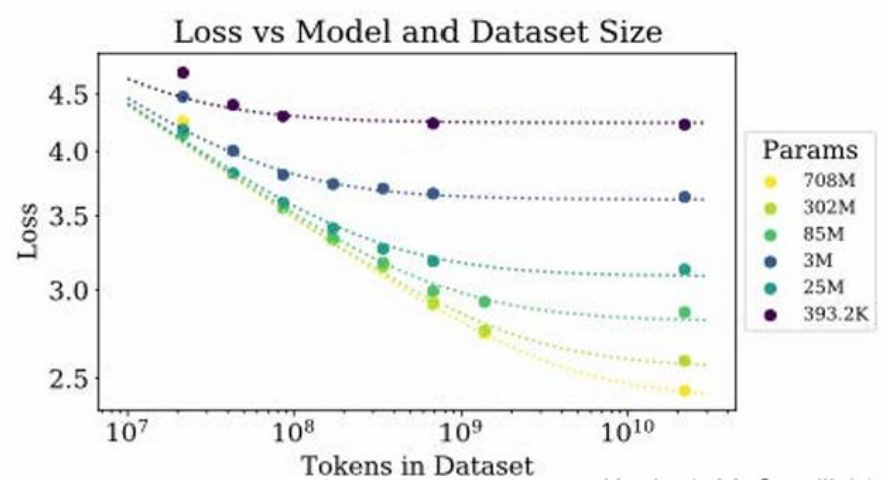
Pick minimum loss at each amount of compute

Run linear regression
on the resulting (loss, compute) pairs

Simultaneous dependance on Data and Model size

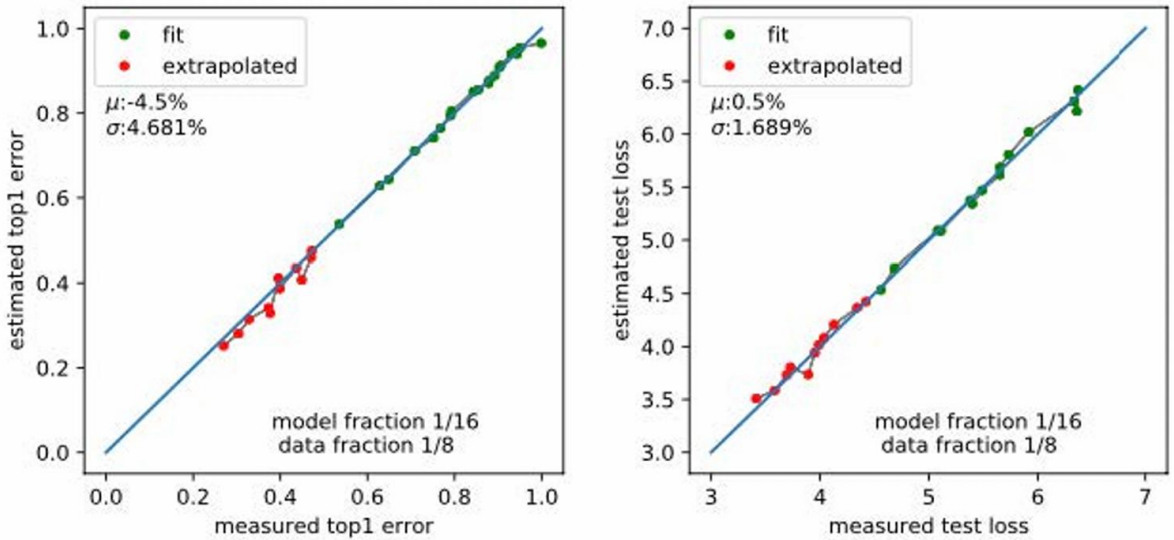
$$N_{opt}(C), D_{opt}(C) = \underset{N,D \text{ s.t. } \text{FLOPs}(N,D)=C}{\operatorname{argmin}} L(N, D)$$

$$L(N, D) = \left[\left(\frac{N_c}{N}\right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D}\right]^{\alpha_D}$$



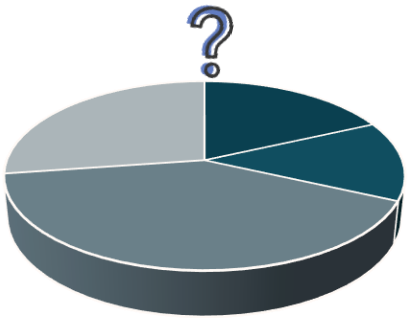
Extrapolation is possible!

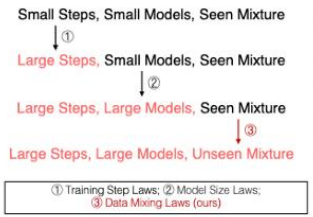
Fit at some scale, make valid predictions at another scale

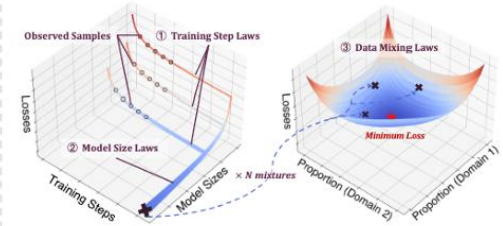


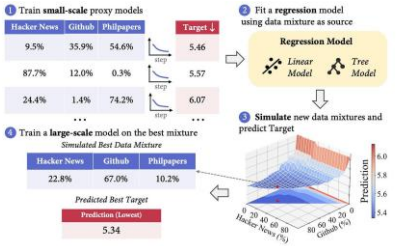
Other data scaling laws: pretraining data optimization

Natural idea – build data scaling laws:

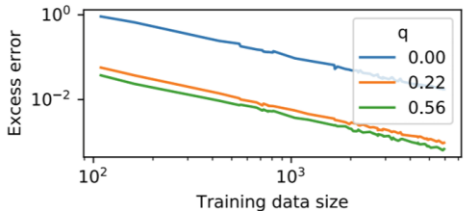


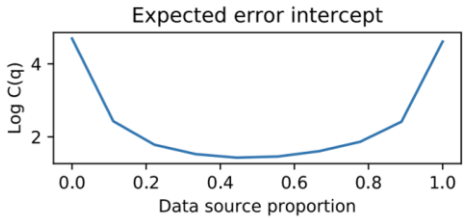






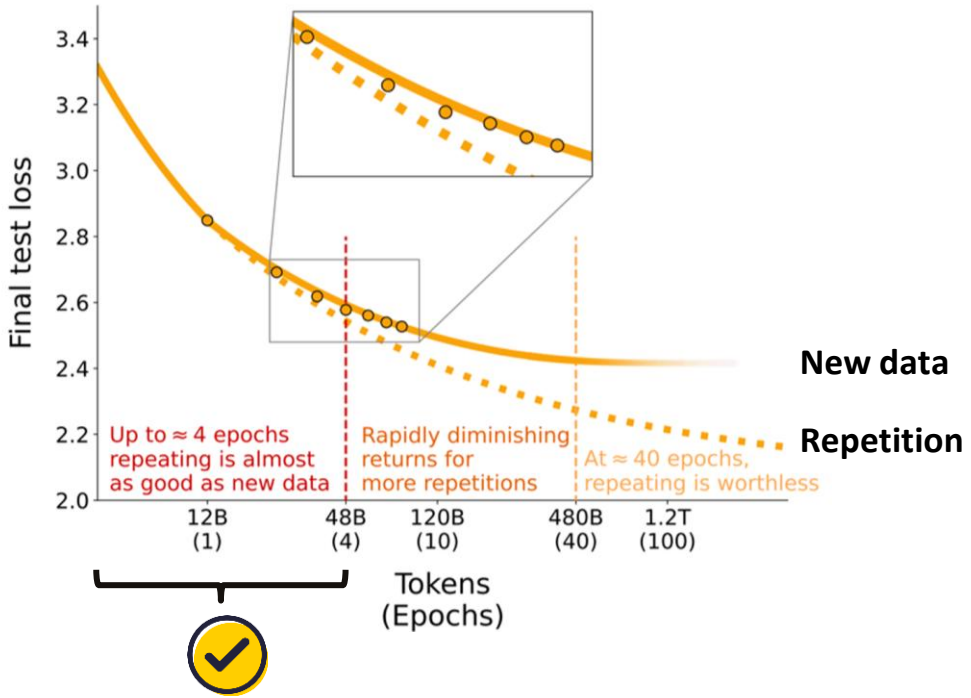
Empirical eval – just take the best small dataset (session 8):





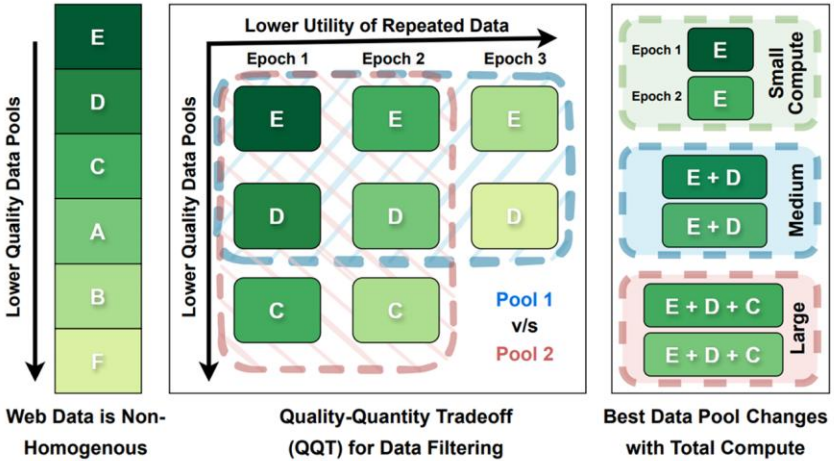
Other data scaling laws: data repetition

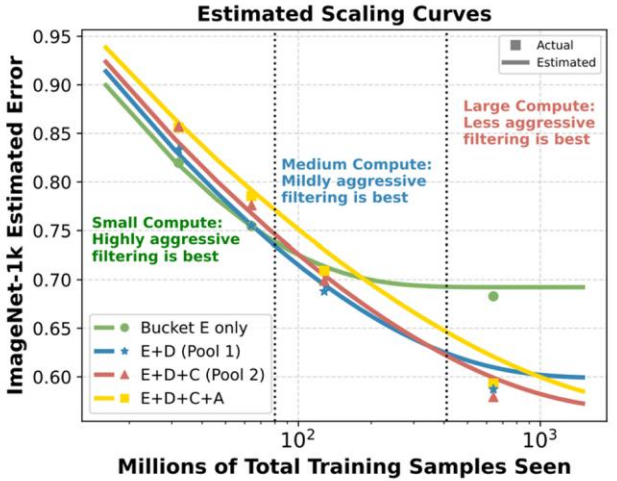
We have finite data → let's think about repetition



Other data scaling laws: data filtering

- More compute → less repetition → looser filter
- The optimal balance between data quality and quantity changes as we scale.



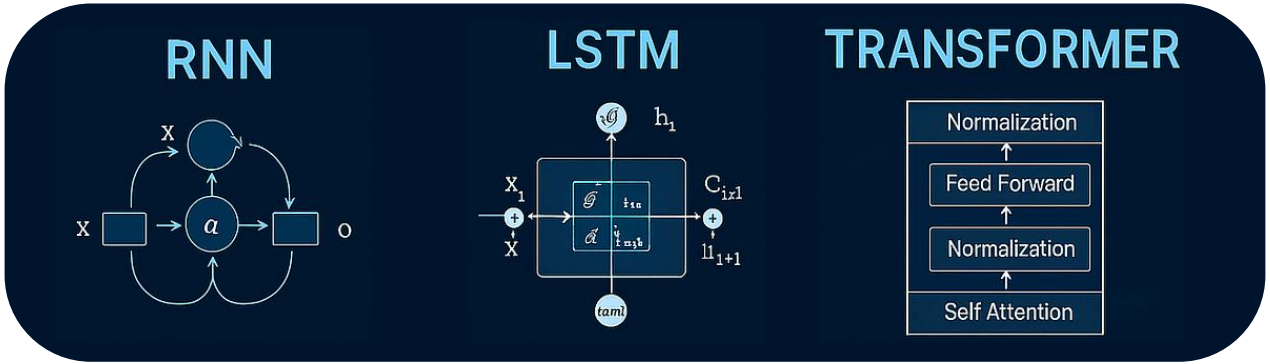


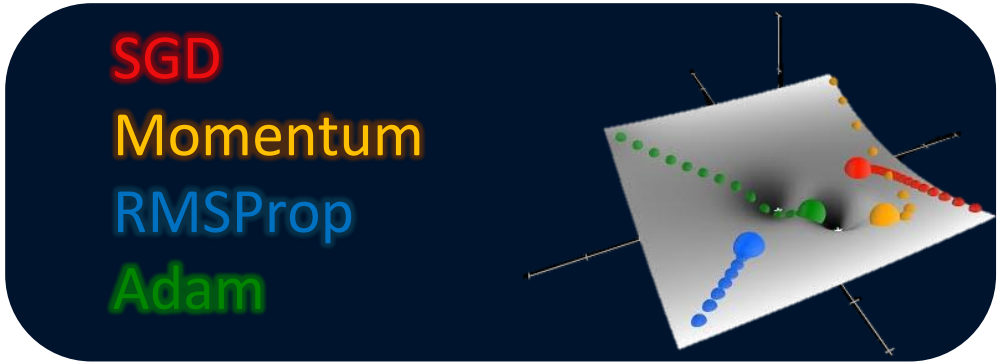
3.

Scaling laws for model engineering

Main Resource For this Section: https://cs336.stanford.edu/ and https://cmu-l3.github.io/anlp-spring2026

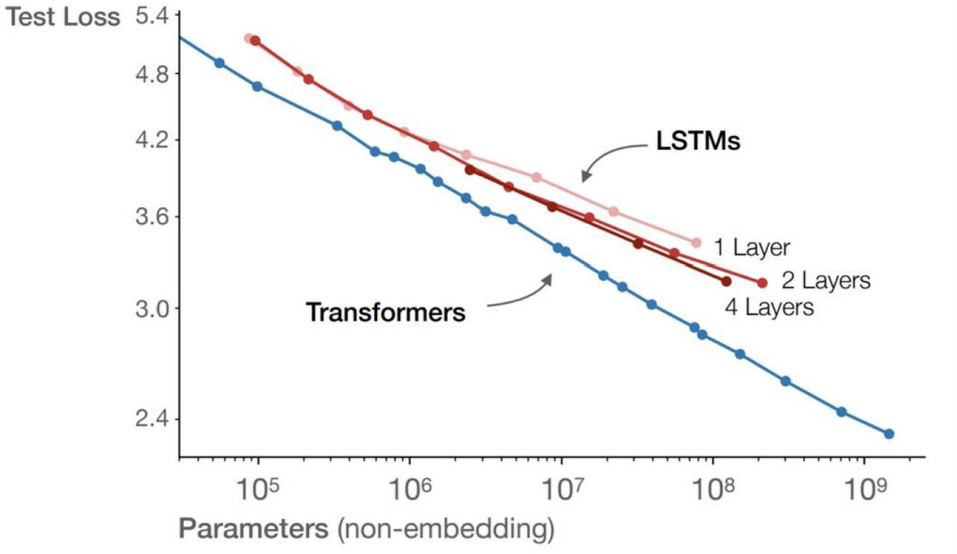
Model scaling law





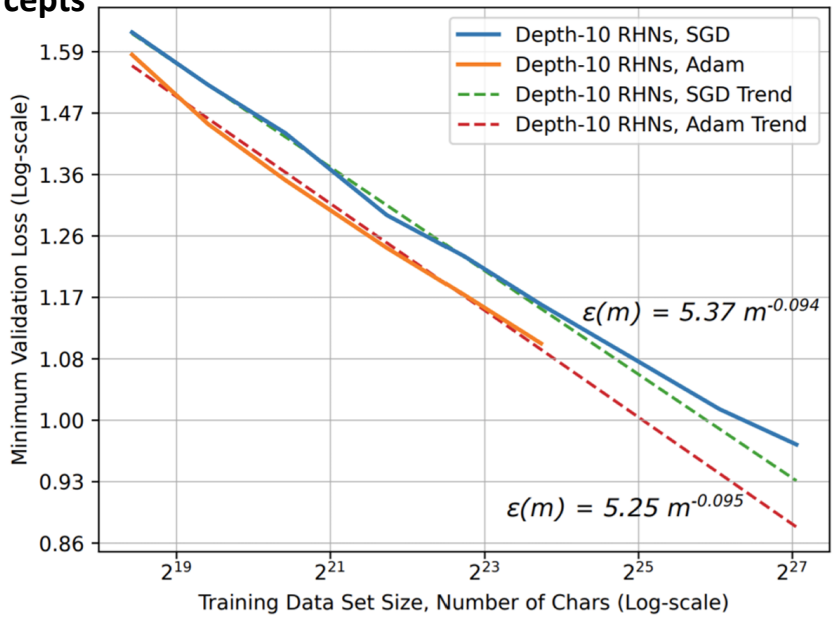
Transformers vs LSTMs: which one's actually better?

Spend tens of millions to train a LSTM GPT-3, or …



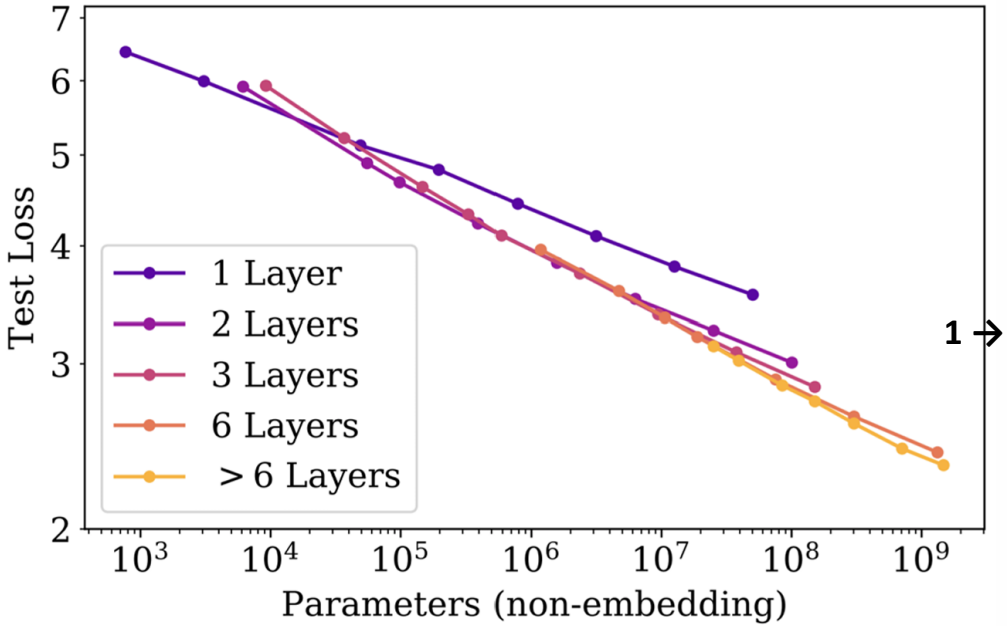
SGD vs Adam

Different intercepts



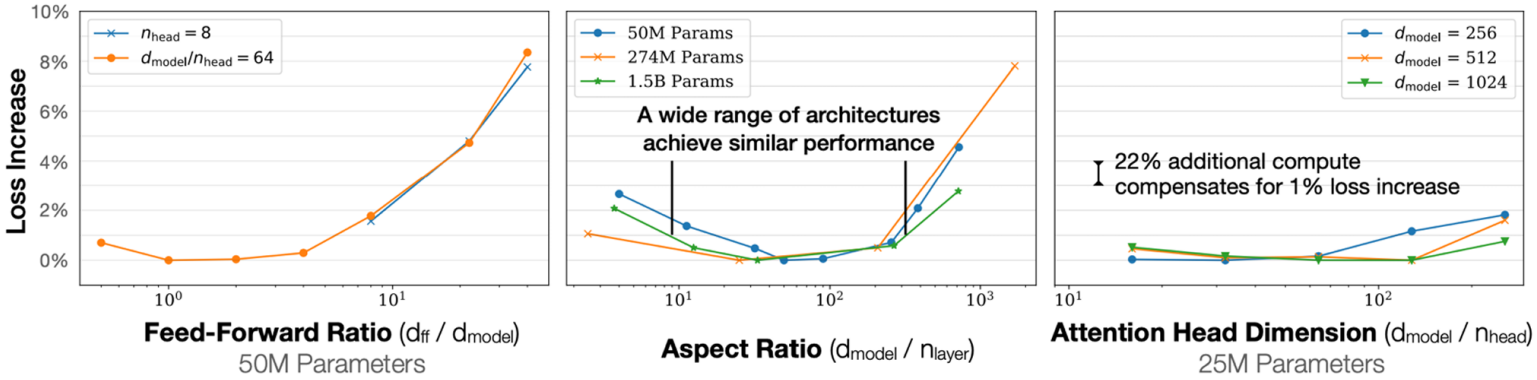
But similar slopes!

Number of layers

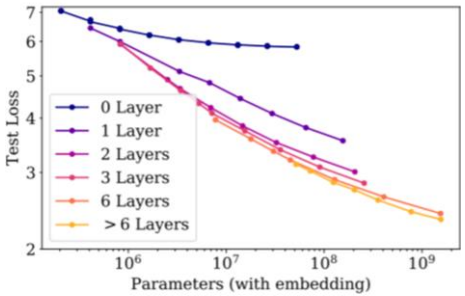


1 → 2 layers: huge difference

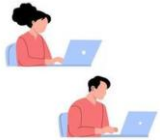
Some transformers ratios



Scaling laws don't just appear, we create them







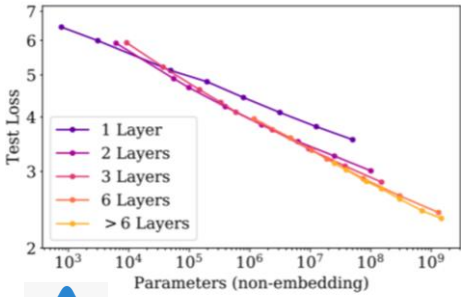
How to get predictable scaling?



Exclude embedding parameters,
They are not doing much computation



Predictability across scales is engineered





Pick the right x-axes

4.

In Context Learning

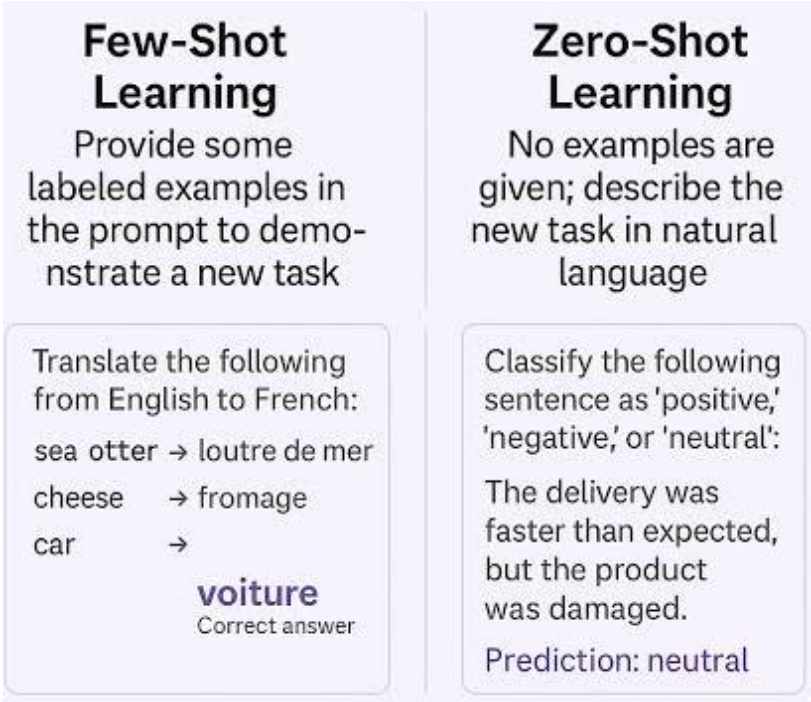
Main Resource For this Section: [Agarwal et al., 2024], [Mehta et al., 2025], https://cs336.stanford.edu/ and https://cmu-l3.github.io/anlp-spring2026

Prompting vs Learning

Few-Shot Prompting

What it is:
You give a few examples (2–5) to teach the task.

Use when:
The task is more complex or creative.



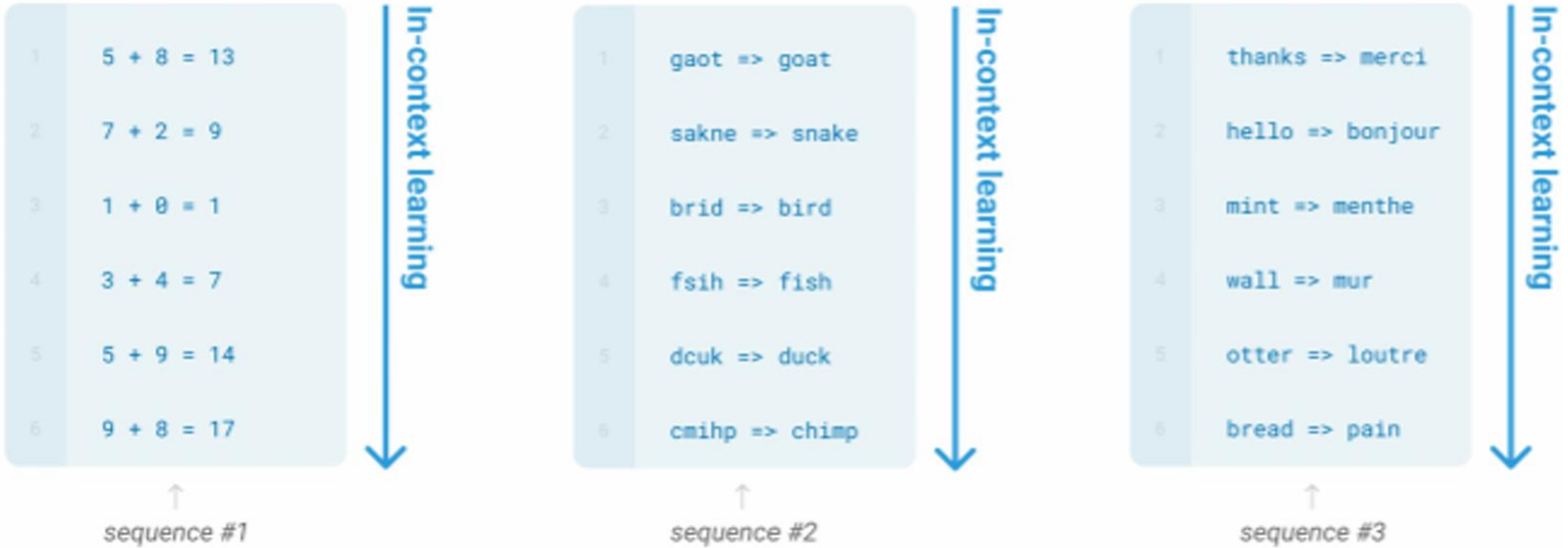
Zero-Shot Prompting

What it is:
You ask the AI to do something without showing any example.

Use when:
The task is simple and clear.

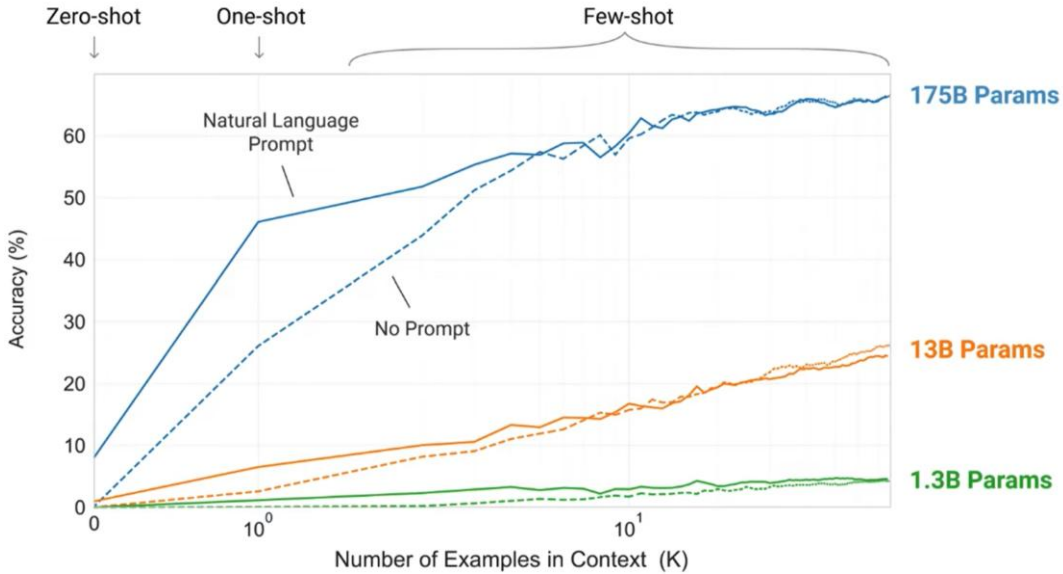
what we provide vs what the model may does with that information

Detect a pattern and continue that pattern

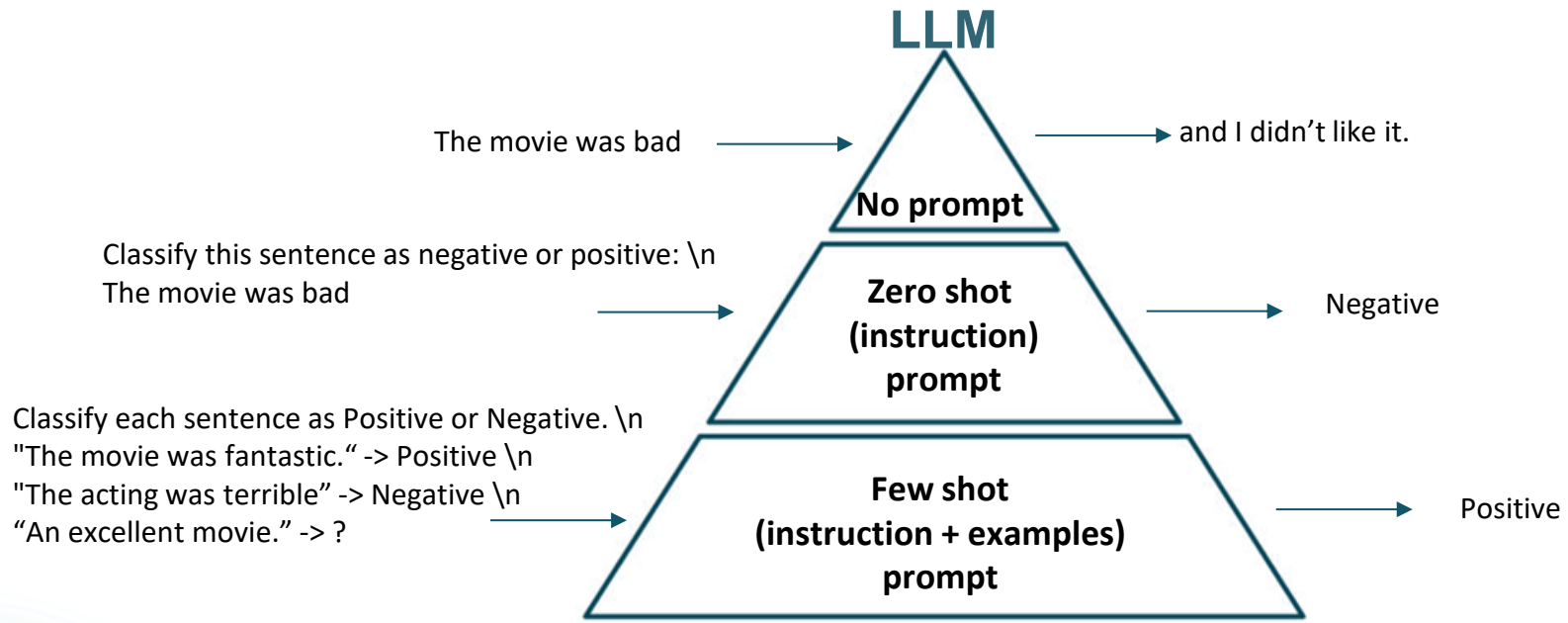


In context learning (ICL) refers to the capability of LLMs to adapt to new tasks and generate correct outputs based solely on the text provided in the inference context window, **without any gradient updates** or weight modifications.

The power of prompting



Communicating with the LLM



Hold on…

Q1. The model has never seen examples of the intended behavior during training?
A: Not necessarily! It might already be familiar with the task from pretraining.

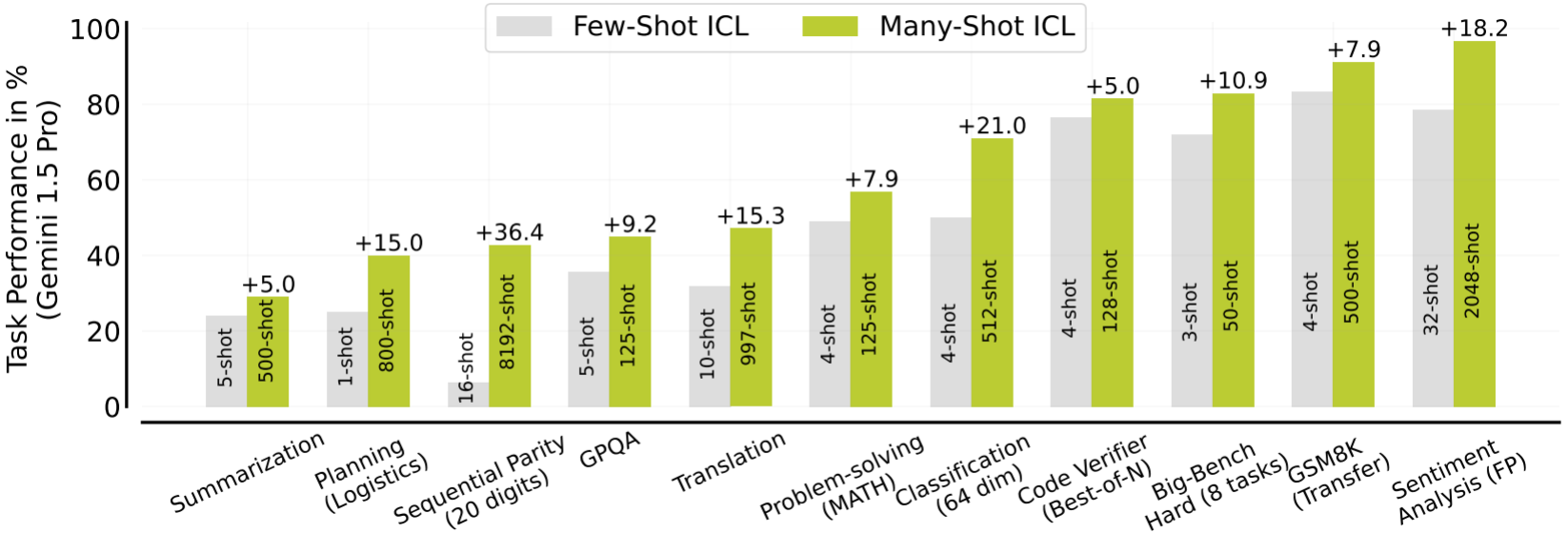
Q2. Then… is it actually learning something new, or just remembering?
A: Both are possible! It may recognize an existing skill or infer a new pattern from context.

Q3. But how can we call it learning if the model's weights don't change?
A: Because adaptation can happen through context, without changing parameters. Whether that constitutes learning depends on how we define it.

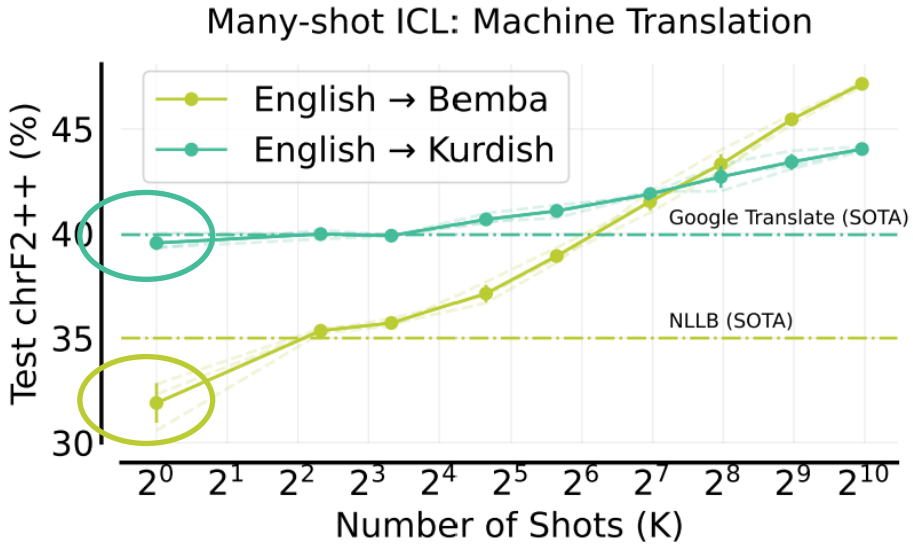
Q4. So… what exactly is happening inside the Transformer during ICL?
A: That's the interesting part! Researchers have proposed several explanations, including mechanisms resembling gradient descent.



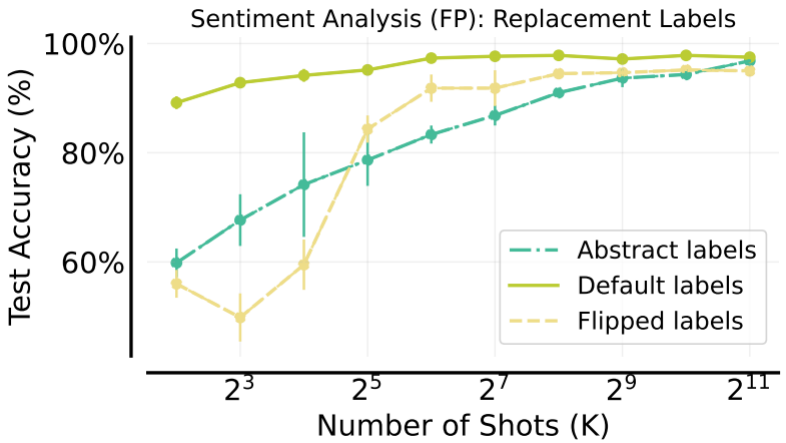
From few-shot to many-shot



Machine translation



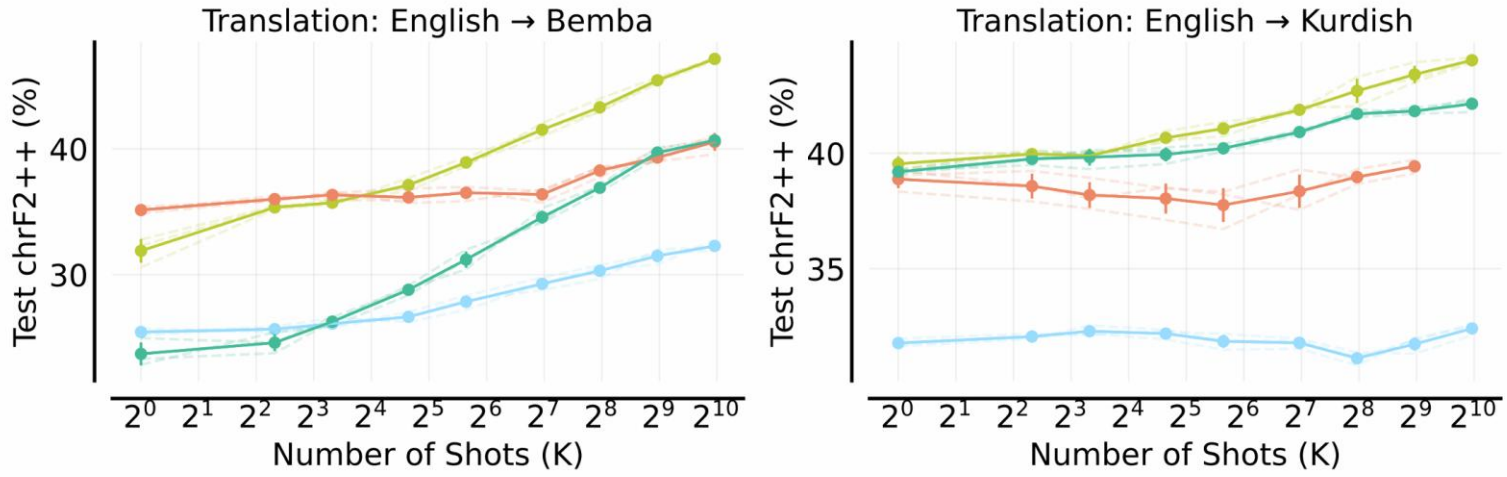
Tricking the model



Flipped labels: [‘negative’, ‘neutral’, ‘positive’] → [‘neutral’, ‘positive’, ‘negative’]

Abstract labels: use [‘A’, ‘B’, ‘C’], removing any pre-existing sentiment association

Ability to leverage many examples varies by model

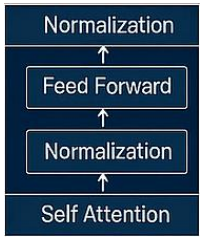


In context “learning”?

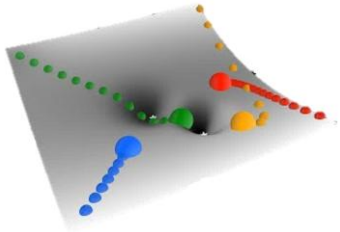
Scaling Laws and In-Context Learning: A Unified Theoretical Framework

Sushant Mehta
sushant0523@gmail.com

Ishan Gupta
345ishaan@gmail.com



≈



Under certain assumptions,
transformer can implement gradient descent through its forward pass
(Attention output can resemble the gradient update)

Scaling of ICL

Model, Data → $\epsilon \propto (ND)^{-\alpha}$

$$\epsilon(N, D, k, n) \leq C \left[\left(\frac{N_0}{N} \right)^{\alpha} + \left(\frac{D_0}{D} \right)^{\alpha} + \left(\frac{k_0}{k} \right)^{\gamma} + \left(\frac{n_0}{n} \right)^{\delta} \right]$$

where $\alpha = \frac{1}{2(h+1)}$, $\gamma = \frac{\beta}{2\beta + d_x}$, $\delta = \frac{1}{2}$, *and* C, N_0, D_0, k_0, n_0 *depend on architecture.*

Theoretical vs measured: Excellent agreement

Task Type	Hierarchy h	α_{theory}	α_{measured}
Linear Regression	0	0.50	0.48 ± 0.02
Sparse Linear	1	0.33	0.31 ± 0.03
Decision Tree	2	0.33	0.32 ± 0.03
Decision Tree	3	0.25	0.23 ± 0.02
Decision Tree	4	0.20	0.19 ± 0.03

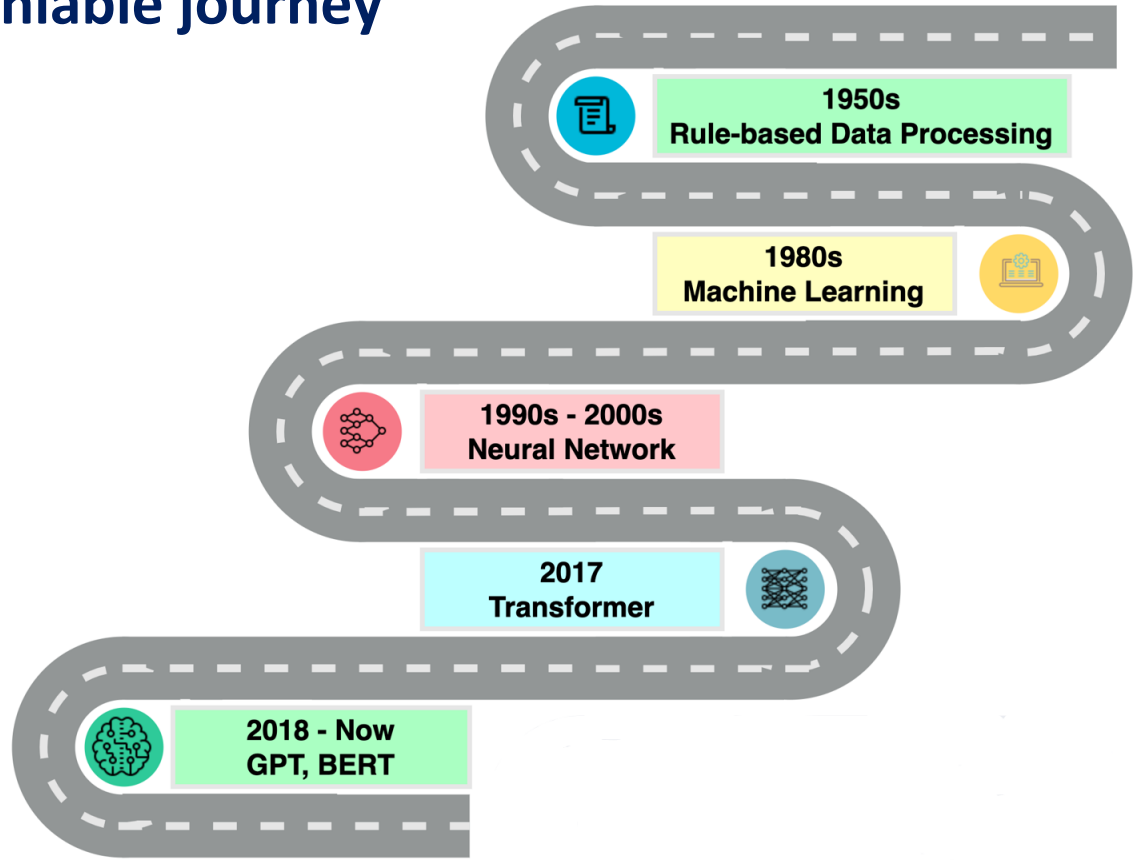
$x \rightarrow 2x$ **VS** $if\ x > 5: x \rightarrow 2x$ else: … **VS** *if* / \ / \

5.

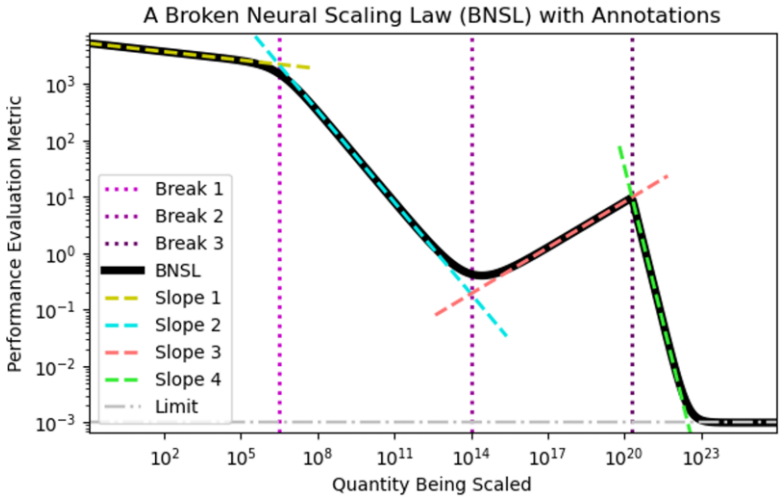
Is Scaling is all we need?

Main Resource For this Section: [Caballero et al., 2023]

Undeniable journey

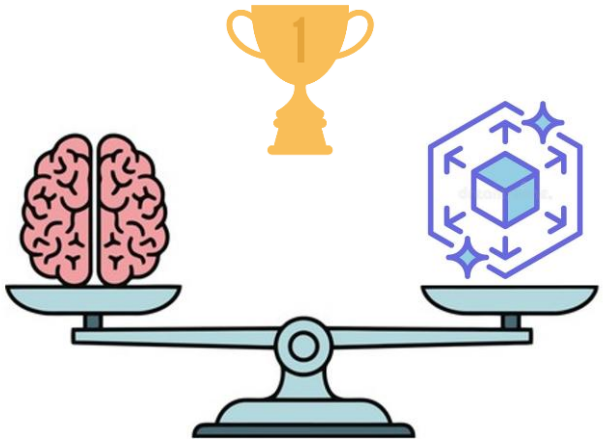


When Scaling laws break down, but The War Ends





Clever Ideas vs Simple Ideas that Scale Well



Advanced NLP - Session 9 – Scaling laws & In-Context Learning - IUST

Scaling is all you need

Scaling laws for data and model size

Scaling laws for model engineering

In-Context learning

Is scaling all we need?